



Education

Lab 5.1. Installing Argo Rollouts

Objective

Set up a local Kubernetes cluster using Kind, install Argo Rollouts, and understand how to access Argo Rollout resources.

Prerequisites

- Basic understanding of Docker, Kubernetes, and command-line interface operations.
- Access to a computer with an internet connection.
- An installation of Kubernetes that you have full control over
 - See Chapter 2's Deploying Kubernetes for Argo section for details on how to set one up for yourself

Install Cluster and Argo Rollouts

NOTE: Steps 1-4 might not be necessary if you already followed the setup during a previous chapter.

1. Installing Docker

Ensure Docker is installed and running on your machine.

- Installation instructions can be found on the [Docker website](#).

2. Installing Kind

Download and install Kind following the instructions from the [Kind official website](#).

3. Creating a Kubernetes Cluster with Kind

Create a cluster by running the following command:

```
kind create cluster
```

This command creates a single-node Kubernetes cluster running inside a Docker container named **kind-kind**.

4. Installing kubectl

Instructions for downloading kubectl can be found in the [Kubernetes official documentation](#).

5. Deploy Argo Rollouts

Create a namespace for Argo Rollouts using the following command:

```
kubectl create namespace argo-rollouts
```

Deploy Argo Rollouts using the quick start manifest:

```
kubectl apply -n argo-rollouts -f  
https://github.com/argoproj/argo-rollouts/releases/download/v1.8.3/install.yaml
```

This will install custom resource definitions as well as the Argo Rollouts controller.

During this course we use Argo Rollouts in version 1.8.3. We recommend using the same version to ensure consistent results.

Verify that Argo Rollouts is installed by running the following command:

```
kubectl get pods -n argo-rollouts
```

6. Install Rollouts kubectl Plugin

Unlike Argo CD and Argo Workflows, Argo Rollouts uses a kubectl plugin as its CLI client.

Download the latest Argo Rollouts kubectl plugin version from <https://github.com/argoproj/argo-rollouts/releases/latest/>.

On Ubuntu 24.04, you can install the CLI using the following commands:

```
$ wget
https://github.com/argoproj/argo-rollouts/releases/download/v1.8.3/kub
ectl-argo-rollouts-linux-amd64 -O kubectl-argo-rollouts

$ chmod +x kubectl-argo-rollouts

$ sudo mv kubectl-argo-rollouts /usr/local/bin/
```

More detailed installation instructions can be found via the [CLI installation documentation](#).

This is also available in Mac, Linux and WSL Homebrew. Use the following command:

```
brew install argoproj/tap/kubectl-argo-rollouts
```

Verify that the `argo` CLI is installed correctly by running the following command:

```
kubectl argo rollouts version
```

7. UI Dashboard

For the sake of completeness it needs to be mentioned that Argo Rollouts ships with a fully fledged UI Dashboard. It can be accessed via the `kubectl argo rollouts dashboard` command and provides a nice overview and basic commands for administration.

```
$ kubectl argo rollouts dashboard
```

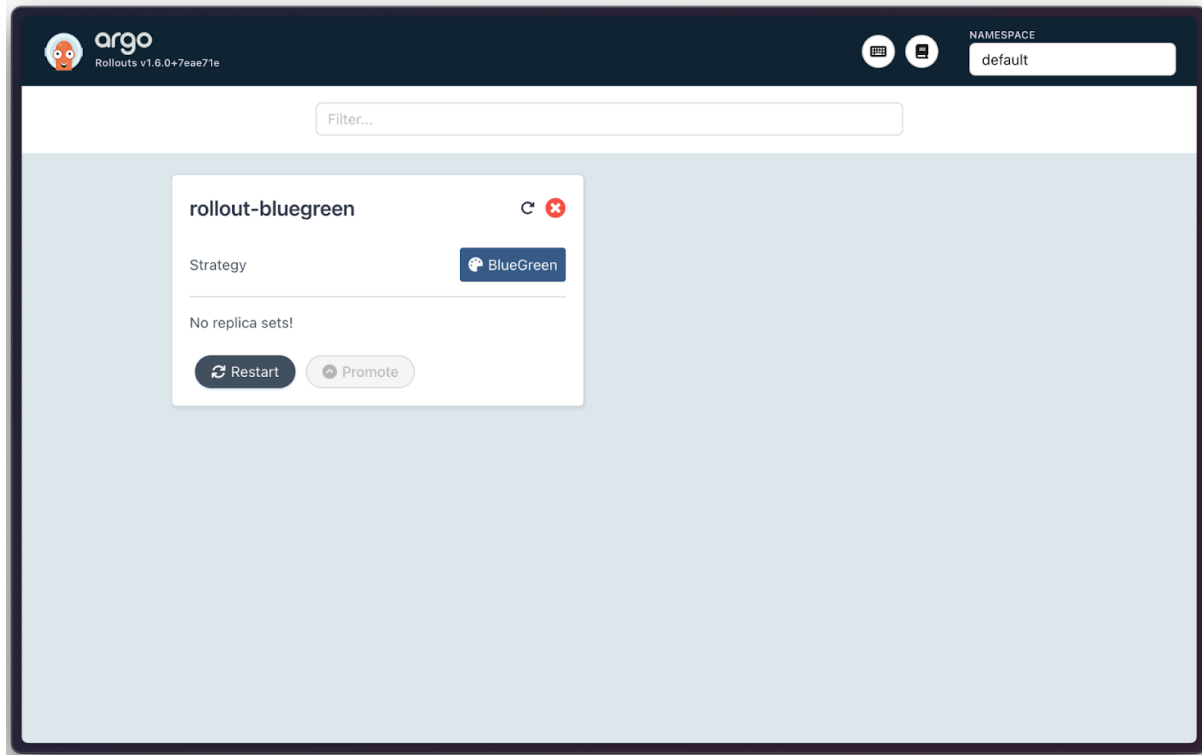
Output:

```
INFO[0000] Argo Rollouts Dashboard is now available at
http://localhost:3100/rollouts

INFO[0000] [core] [Channel #1 SubChannel #2]grpc:
addrConn.createTransport failed to connect to {Addr: "0.0.0.0:3100",
ServerName: "0.0.0.0:3100", }. Err: connection error: desc =
"transport: Error while dialing: dial tcp 0.0.0.0:3100: connect:
connection refused"
```

Despite any “connection refused” errors, you can now access it via the UI at `http://localhost:3100` or your VM’s public IP address at port 3100.

As the Dashboard is self-explanatory, we will not discuss it in detail during this course.



The Argo Rollouts Dashboard displaying a sample app “rollout-bluegreen”

NOTE: If no rollout resources are in place, the dashboard will display “Loading...”.

8. Optional: Shell Auto-Completion

To get easy access to Argo Rollout resources, the CLI can add shell completion code for several shells. For bash, you can use the following command:

```
source <(kubectl-argo-rollouts completion bash)
```

Other shells are supported as well. Please refer to the [completion command documentation](#) for more details.

9. Using Argo Rollouts

There are a wide variety of commands that you can use to control Argo Rollouts via the CLI, as described in the `-h` output for the `kubectl argo rollouts` command. As a `kubectl` plugin, it uses the Kubernetes API to perform all management tasks.

Here is a list of the most common commands to operate with Argo Rollouts:

```
kubectl get rollout
```

```
kubectl argo rollouts get rollout
```

```
kubectl argo rollouts promote
```

```
kubectl argo rollouts undo
```